

Day 3 :- RAM and Storage Theory : types of RAM(DDR3, DDR4,DDR5) and storage (HDD vs. SSD, NVMe).

Ans. :-

Tasks 1 :-Create a comparison table listing the main differences between DDR3, DDR4, and DDR5 RAM, including speed, voltage, and typical use cases.

Feature	DDR3	DDR4	DDR5
Release Year	2007	2014	2020
Standard Speed Range	800–2133 MT/s	1600–3200 MT/s	4800–8400+ MT/s
Typical Consumer Speeds	1333–1866 MT/s	2400–3200 MT/s	5200–7200 MT/s
Bandwidth	Lowest	Higher than DDR3	Highest
Capacity per Module	16 GB	64 GB	Commonly 32–128 GB+
Power Efficiency	Least efficient	More efficient than DDR3	Most efficient
Physical Compatibility	Not compatible with DDR4/DDR5 slots	Not compatible with DDR3/DDR5 slots	Not compatible with DDR3/DDR4 slots
Operating Voltage	1.5 V (1.35 V for DDR3L)	1.2 V	1.1 V

Tasks 2 :- Open your laptop or desktop (with power disconnected), locate the RAM slots, and take a clear photo of the installed RAM module. Reinstall the RAM module and verify that your system boots up successfully.

Hint: Refer to your device manual or a trusted YouTube guide if unsure about RAM removal and installation steps.



Steps to Install a RAM Module :-

1. Shut down the computer completely.
2. Disconnect the power cable and remove the battery (if applicable).
3. Open the computer case or laptop service panel.
4. Locate the RAM slots on the motherboard.
5. If replacing existing RAM, release the locking clips and remove the old module.
6. Align the notch on the RAM module with the notch in the RAM slot.
7. Insert the RAM module into the slot at the correct angle.
8. Press down firmly until the locking clips click into place.
9. Ensure the RAM module is securely seated and the clips are locked.
10. Close the computer case or service panel.
11. Reconnect the power cable and battery.
12. Turn on the computer and verify that the system detects the installed RAM correctly.

Tasks 3 :- List three key differences between HDD, SSD, and NVMe storage types, and for each, name one app or use case (from apps you use daily) that would benefit most from that storage type.

Feature	HDD(Hard Disk Drive)	SSD(solid State Drive)	NVMe SSD
Storage Technology	Uses spinning magnetic disks and a moving read/write head	Uses flash memory with no moving parts	Uses flash memory and connects through the PCIe interface
Speed	Slowest storage type	Much faster than HDD	Fastest storage type available for consumer PCs
Durability	Less durable because of moving parts	More durable due to no moving parts	Highly durable and reliable
Boot Time	Long boot time	Faster boot time	Extremely fast boot time
Cost	Lowest cost per GB	Higher cost than HDD	Highest cost among the three

- **Daily APP/ Use Case Examples & it Benefits :-**

- 1. HDD(Hard Disk Drive) :-**

Daily APP/ Use Case :- Movies, Photos, Music Collection , Backup Files

Benefits :- Large Storage Capacity at a low Cost

- 2. SSD Drive :-**

Daily APP/ Use Case :- Windows 11, Spotify, Google Chrome , MS Office

Benefit :- Faster boot time and quicker app loading

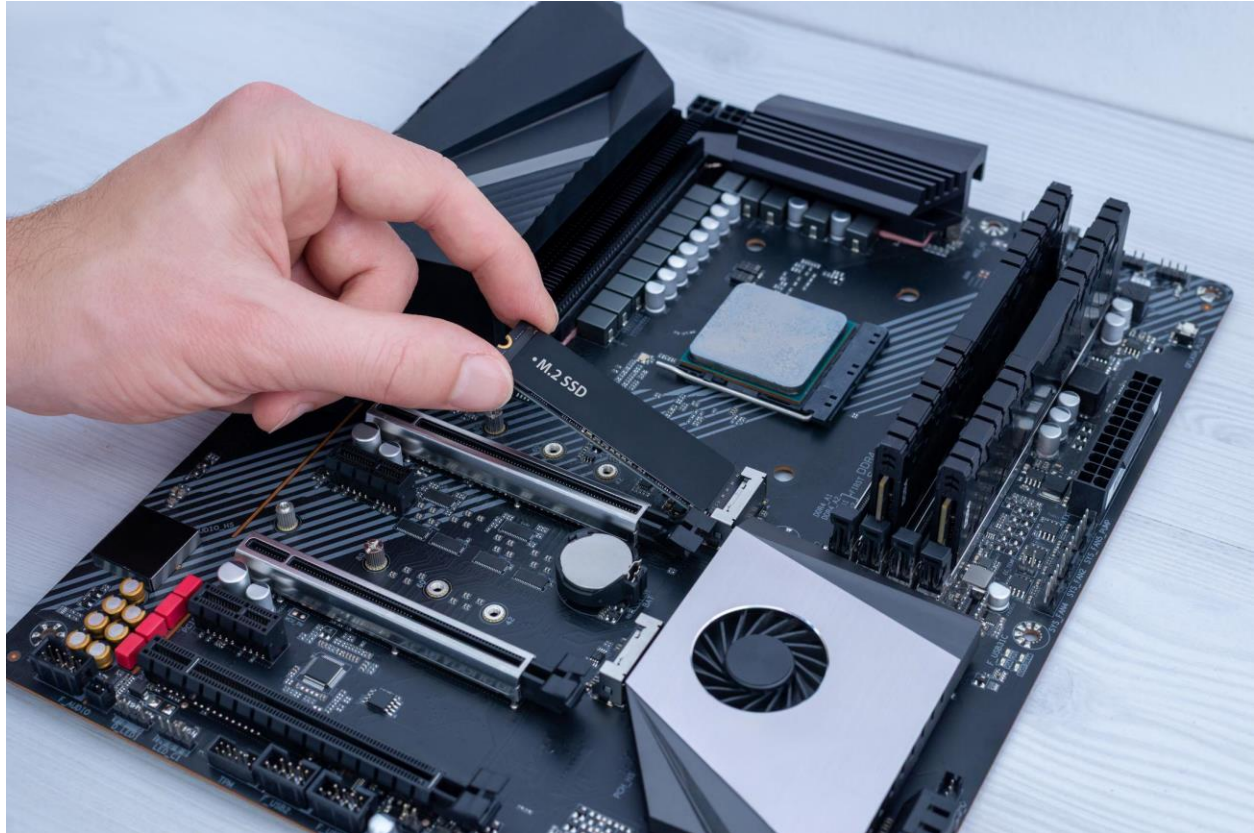
- 3. NVMe SSD :-**

Daily APP/ Use Case :- GTA 5, Adobe Photoshop, Adobe Premiere Pro , Android Studio

Benefit :- Extremely fast loading , better gaming and editing performance

Tasks 4 :- Remove the storage drive (HDD or SSD) from your system, take a photo of the connector type (SATA, M.2, or NVMe), and reinstall the drive. Confirm that your device recognizes the storage after reassembly.
Constraint: Only perform this task if

you have access to a device you are allowed to open. If not, find and share a YouTube video link showing the process for your laptop/PC model.



If you **have access to a device you are allowed to open**, follow these steps :-

1. Shut down the computer and disconnect power.
2. Open the laptop/desktop case.
3. Locate the storage drive:
 - **SATA HDD/SSD** – connected by a SATA data cable and power cable.
 - **M.2 SATA SSD** – small card mounted directly on the motherboard.
 - **M.2 NVMe SSD** – similar to M.2 SATA but uses the NVMe interface.
4. Take a clear photo showing the connector type.
5. Carefully remove the drive.
6. Reinstall it securely.
7. Reassemble the system and power it on.
8. Verify the drive is detected:

9. **Windows:** Open *This PC* or *Disk Management*.
 10. **Linux:** Run `lsblk` or check the file manager.
 11. Take a screenshot showing the drive is recognized.
- [How to Remove/Install a M.2 SSD \(YouTube\)](#) — Shows how to locate, remove, and reinstall an M.2/NVMe SSD.